PRODUCT PERFORMACE UPDATE

Application: Nucor Gallatin Steel, Tunnel Furnace

FURNACE DETAILS

- **Lining Configuration:** The tunnel furnace is 680 ft. in length with ceramic fiber upper sidewalls and roof and hard cast lower sidewalls and hearth. The upper sidewalls and roof are stack bonded fiber construction with 9" of insulation on the sidewalls and 12" on the roof.

- **Customer Need:** ITC recommended a solution to address several major challenges,
  - Heat loss and high shell temperatures on the roof and sidewalls due to shrinking fiber at operating temperatures.
  - Frequent refractory maintenance to both fiber and hard cast insulation.

ITC RECOMMENDATION

- **Engineered Approach:** ITC recommended applying a proprietary high temperature, energy efficient ceramic coating on the hot face of the ceramic fiber insulation to eliminate shrinking of the ceramic fiber and hot spots on the furnace shell. The project consisted of coating all 240 linear feet of newly installed fiber on both the upper sidewalls and roof.

- **High Temperature Energy Efficient Ceramic Coatings:** ITC manufactures a proprietary, high temperature, energy efficient ceramic coating to work in concert with the furnace conditions and all types of refractory design. Under ideal conditions, high temperature, energy efficient ceramic coatings absorb energy from the process and re-radiate it to the furnace load, imparting the following benefits…

  Lower heat loss – Minimizes heat transfer through the furnace refractory walls
  Lower Maintenance costs – Protects the substrate refractory
  Improved furnace operation – Improves the thermal efficiency of the process

Newly Installed Fiber

Coated Roof Sections

IMPACT ON THE PROCESS

- LOWER SHELL TEMEATURES
  The combination of new fiber and high temperature energy efficient ceramic coatings to protect the fiber from shrinking and devitrification, have lowered shell temperatures and hot spots.

- IMPROVED BTU SAVINGS
  Due to the re-radiating properties of the coating, Btu savings are significantly better than uncoated fiber refractory.

- REDUCED REFRACTORY MAINTENANCE

- INCREASED REFRACTORY LONGEVITY